

Nomenclator Key Reconstructor — User Guide

Table of Contents

- [Nomenclator Key Reconstructor — User Guide](#)
 - [Table of Contents](#)
 - [Introduction](#)
 - [Core Concepts](#)
 - [Tokens](#)
 - [Plaintext and Ciphertext](#)
 - [NULL Tokens](#)
 - [Getting Started](#)
 - [Step 1: Open the app](#)
 - [Step 2: Choose Parse Mode](#)
 - [Quick Shortcuts and Actions](#)
 - [Parsing Modes](#)
 - [Separator Mode](#)
 - [Fixed-Length Mode](#)
 - [Analysis & Frequency Scoring](#)
 - [Mapping Modes](#)
 - [Single-key Mapping](#)
 - [Homophone \(Multi-key\) Mapping](#)
 - [Mapping Controls](#)
 - [Shift Controls \(Fixed-Length\)](#)
 - [Extract and Reattach NULL Cells](#)
 - [Lock and Unlock Mappings](#)
 - [Suggestions \(Soft Mode\)](#)
 - [Quick Assign](#)
 - [Merge PT / Swap CT Tokens / Edit Cells](#)
 - [Review & Export](#)
 - [Review Tentative Mappings](#)
 - [Resolve Mapping Conflicts](#)
 - [Mark Null \(Deception\) Tokens](#)

- [Export Reconstructed Key](#)
- [Examples](#)
 - [Example 1: Separator mode, single-key, with deception token](#)
 - [Example 2: Fixed-length with shift and NULL extraction](#)
 - [Example 3: Homophone mode](#)
- [Troubleshooting](#)
 - [Issue: "Shift Left/Right" is greyed out.](#)
 - [Issue: Parsing fails or tokens are unexpectedly split.](#)
 - [Issue: Homophone cells are marked "tentative".](#)
 - [Issue: Suggestions are missing \(or appear unexpectedly\).](#)
 - [Issue: Shift is disabled even though I need to rebalance groups.](#)
 - [Issue: Export contains unexpected NULL tokens.](#)
 - [Issue not known](#)
- [Appendix: Glossary](#)

Introduction

Nomenclator Key Reconstructor is a tool for cryptanalysts to reconstruct substitution cipher keys (nomenclators) through frequency analysis, manual mapping, and interactive exploration. The tool supports both **fixed-length** and **separator-based** parsing of ciphertext and also **homophone (multi-key)** handling.

This guide explains the core concepts, workflows, and controls.

Core Concepts

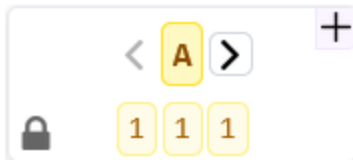
Tokens

A **token** is the atomic unit of ciphertext that maps to a plaintext character:

- **In Separator mode** (e.g., separator = `:`):
 - Ciphertext `"A:111:B:22"` is tokenized as `['A', '111', 'B', '22']`
 - Each separator-delimited part is a single token.
 - A cell with 1 token holds one complete separator-delimited group.
 - Example token here is `['111']`



- **In Fixed-Length mode** (e.g., group size = 3):
 - Ciphertext "ABC111B22" is tokenized as ['1', '1', '1', 'A', 'B', 'C', 'B', '2', '2']
 - Each character is a single token.
 - The mapping grid groups tokens into "cells" (bundles of up to `maxLen` tokens).
 - Example: a cell with 3 tokens in fixed-length mode might visually represent ['1', '1', '1']. In this mode, you can extract edge tokens and reattach them (effectively changing token grouping at the mapping level). **THIS IS NOT POSSIBLE IN SEPARATOR MODE.**



TODO How to explain tokens better???

Plaintext and Ciphertext

- **Plaintext (PT):** The original message. User can type it or import it.
- **Ciphertext (CT):** The encrypted message. User can type it or import it. Parsed and tokenized, then mapped to PT characters.
- **Mapping Grid:** An interactive table where each row contains PT characters. Each PT cell is associated with one or more CT tokens, depending on the mode.

NULL Tokens

In a nomenclator cipher, extra tokens are often included to **confuse and deceive** the reader. These tokens:

- Have no corresponding plaintext character.
- Are marked as **NULL** (or "deception") in the tool.
- Can be tagged during review and exported as part of the final key.
- May be present naturally in the ciphertext, or created during mapping (via extract/reabsorb in fixed-length mode).

Getting Started

Step 1: Open the app

1. **Enter Plaintext:**
 - Paste or load the plaintext.
2. **Enter Ciphertext:**
 - Paste or load the ciphertext.

Enter the plain text and cipher text, then configure parsing and run the analysis.

Plain text (PT)
Use [brackets] for multi-char tokens, e.g. [HELLO]WORLD

Import PT

AHAHO

Cipher text (CT)
Tokens separated by space, or a single continuous string, e.g. 11:22:33:33:44

Import CT

99:11:22:11:33:99:44:99

⚠ Warning: PT (5) < CT tokens (8). Choose deception tokens.

Step 2: Choose Parse Mode

Before parsing, select your **parse mode**:

- **Fixed-Length:** Tokens are fixed-width groups (e.g., every 2 characters = 1 token).
 - In this mode, token grouping can later be adjusted by extracting/reattaching edge tokens.
- **Separator:** Tokens are delimited by a separator character (e.g., : or space).

PARSING CONFIGURATION

[What is this for??](#)

CT format: Separated by character ▼ Separator: :

Keys per PT char: Multiple (homophones) ▼

Reset

Run analysis

Then click **Run analysis** to generate candidate mappings.

Quick Shortcuts and Actions

- **Frequency modal**: open it with the **Frequency** button in the navbar, or press **Shift**. This keyboard shortcut opens the frequency of **PT** and **CT**;
- **Clear saved data**: click **Clear data** in the navbar, or press **Esc**.
- **Important**: Clear data asks for confirmation and then removes saved local data.

Parsing Modes

Separator Mode

Use case: Ciphertext where tokens are delimited by a specific character (e.g., spaces or colons).

How it works:

- Specify the **separator** (e.g., : or space).
- The ciphertext is split by this separator into discrete tokens.
- Example: "A:11:B:22" with separator : → tokens ['A', '11', 'B', '22'] .

Mapping:

- Each token is treated as a complete unit and mapped to PT characters.
- In separator mode, each PT cell typically holds one token (though multiple tokens can be grouped if needed).

Fixed-Length Mode

Use case: Ciphertext where each token occupies a fixed number of characters (e.g., 2-character groups).

How it works:

- Specify the **group size** (e.g., 2).
- The ciphertext is first tokenized into individual characters (spaces are ignored).
- Group size controls how many characters are grouped per PT cell during mapping.
- Example: "A11B22" with group size 2 gives character tokens ['A', '1', '1', 'B', '2', '2'], grouped as ['A1', '1B', '22'] for mapping.

Mapping:

- Once tokens are defined, each PT character is mapped to one or more tokens.
- If PT/CT counts are not cleanly aligned, status warnings are shown (`needsNull` or `invalid`).
- The **Shift Controls** allow you to move tokens between PT cells while maintaining a consistent token distribution.
- Shift can also be useful after manual edge extraction, when regrouping was changed.

Analysis & Frequency Scoring

The tool performs **frequency analysis** to suggest candidate mappings:

1. Click **Run analysis**:

- Computes the frequency distribution of each CT token.
- Scores possible mappings based on frequency correlation with known PT character frequencies (if available).

2. **View Candidate Mappings (Suggestion table)**:

- For each PT character, the tool displays candidate CT tokens ranked by score.
- This ranked list is the **analysis candidate list** (available after analysis, even before locks).
- Score used in the code (simple explanation): the analyzer counts how often a ciphertext token appears (`tokenCount`) and how many plaintext cells contain the PT character (`cellCount`). Then:
 - In multi-key (homophone) mode: $\text{score} = \min(\text{tokenCount} / \text{cellCount}, 1.0)$.
 - Otherwise: $\text{score} = \min(\text{tokenCount}, \text{cellCount}) / \max(\text{tokenCount}, \text{cellCount})$.

Both formulas produce values in [0, 1], where higher means better frequency agreement.

- Higher-scored candidates are more likely based on frequency analysis.

Suggestions

Clear all
Preview
✓ Apply

Mode: Single-key

Assigned: 3 / 3

A

11 (score: 1.00) ▼

H

22 (score: 1.00) ▼

O

33 (score: 1.00) ▼

3. Auto-Pick Best Candidates (optional):

- Automatically selects score == 1 candidates.
- In single-key mode, the pick is applied only when it is unique for that PT character.
- If some characters are ambiguous, unambiguous picks are still applied and ambiguity is reported.
- Useful as a starting point; manual refinement may still be needed.
- IMPORTANT** Be careful, in multi-key mode this works only where there is no **HOMOPHONES**.

4. Frequency View (optional but useful):

- Open Frequency via the **Frequency** button or **Shift** shortcut.
- Use it before and after marking deception tokens to quickly verify distribution changes.
- My advice** Open it always in beginning, and sort PT ascendent, and start with those characters.

5. Suggestions (soft guidance):

- These are **lock-derived suggestions** (different from the base analysis candidate list).
- Suggestions are visual hints only; they do not auto-lock mappings.
- Suggestions are shown only after at least one mapping is locked.
- Single-key mode: a suggestion is shown only when exactly one stable token is inferred.
- Multi-key mode: suggested candidate checkboxes are highlighted.
- Ambiguous intervals between locked anchors (containing NULL/deception or tentative cells) do not produce suggestions.



- According to these locks u can see there will be H suggested as 22 because of the second and third A and their lock.

Mapping Modes

Single-key Mapping

Use case: Each plaintext character maps to exactly one ciphertext token (standard substitution cipher).

Workflow:

1. For each PT character, view the list of candidate CT tokens.
2. Click to **Select** a candidate.
3. Optionally, click **Lock** to confirm the mapping (prevents accidental changes).
4. Unmapped CT tokens remain unassigned.

Example:

- PT character 'E' might map to CT token '11' .
- Later, if you find a conflict, you can **Unlock** and select a different candidate.

Homophone (Multi-key) Mapping

Use case: A plaintext character maps to *multiple* ciphertext tokens (homophone/nomenclator cipher).

How it works:

1. Toggle **Homophone Mode** to enable multi-key mapping.
2. For each PT character, you can select and lock *multiple* candidate tokens.

Suggestions

Clear all

Preview

✓ Apply

Mode: Multi-key (homophones)

A

1 homophone

☒ 11

1.00 (2/2)

H

0 homophone

☐ 22

0.50 (1/2)

☐ 44

0.50 (1/2)

O

1 homophone

☒ 33

1.00 (1/1)

3. The tool uses **local lookahead** to find tokens that match locked candidates.

Local Lookahead:

- When a PT character has multiple locked tokens, the tool scans forward in the CT sequence to find matching tokens.
- Practical warning: if your earliest lock is wrong, forward-only lookahead can propagate bad assignments. If results degrade, unlock and correct earlier locks first.
- The lookahead window is bounded by the current excess CT groups (extra CT groups beyond PT count).
- The scan **stops at hard-locked boundaries** — tokens that are locked for another PT character.
- This prevents excessive deception/NULL overallocation.
- **My advice** always choose all homophones, because in large texts where is a lot of null tokens, soft lookahead can make more problems.

Confirmed Hard-Locked Boundaries:

- Locks (confirmed mappings) are treated as hard boundaries in the lookahead scan.
- Selections (temporary choices) remain soft and do not block lookahead.

Tentative Cells:

- If a CT token doesn't match any locked candidate for the current PT character, it is tentatively assigned.
- Such cells are marked as "tentative" to signal manual review (amber highlight).
- As I said before in large texts it can be dangerous!!

Mapping Controls

Shift Controls (Fixed-Length)

Available only in fixed-length mode.

Purpose: Dynamically reallocate CT tokens among PT cells while respecting the maximum group size constraint.

How it works:

1. Each PT cell shows the number of CT tokens currently assigned (up to `maxLen`).
2. Click the **Shift Left** or **Shift Right** arrow to move one token between adjacent cells.
3. The tool ensures no PT cell exceeds `maxLen` tokens; if a cell would overflow, tokens cascade to neighboring cells.

UI Constraints:

- The UI **blocks shifts into an immediately locked neighbor**.
- This prevents accidental collision with confirmed mappings at cell boundaries.

Internal Cascade Behavior:

- When a token overflows, the internal simulation cascades it through subsequent cells to maintain the `maxLen` constraint globally.
- **Note:** The cascade is not limited to locked-only zones; it follows global `maxLen` rules.

Example:

- Suppose `maxLen = 2` and cells currently hold `[2, 1, 1]` tokens respectively.
- Shift right from cell 0 (with 2 tokens) → one token moves to cell 1 → cell 1 now has 2 tokens → result is `[1, 2, 1]` .

Extract and Reattach NULL Cells

Available in fixed-length mode; creates NULL (deception) cells.

Extract CT Group to NULL Cell:

- Drag a CT token and drop it on an active edge strip (left or right edge of neighboring cell).
- A new **NULL cell** is created adjacent to the original cell, holding the extracted token.
- Use this to isolate confusing tokens or create deliberate deception cells.

Reattach NULL CT Group:

- Drag a token from the NULL cell back onto an adjacent active edge strip.
- The NULL cell's token is moved back into an adjacent PT cell.
- Use this to undo an extraction or consolidate tokens.

Lock and Unlock Mappings

- **Lock:** Confirm a mapping for a PT character. Locked mappings are immutable unless explicitly unlocked.
- **Unlock:** Release a lock to allow re-selection or adjustment.
- Locked mappings appear visually distinct (GREEN).
- In the Key pairs table, the highlight button is hidden for locked rows.

Suggestions (Soft Mode)

Suggestions help speed up review but remain non-destructive.

- Suggestions are enabled only when at least one lock exists.
- Suggestions are derived from stable mapping cells (non-tentative, non-deception).
- A candidate is suppressed when it lies inside an ambiguous interval between two locked anchors.
 - Example: if an anchor-to-anchor interval contains NULL/deception or tentative cells, that interval does not emit suggestions.
- Single-key mode:
 - The dropdown is highlighted only when exactly one suggestion exists for that PT character.
- Multi-key mode:
 - Suggested checkboxes are highlighted; users still confirm manually.

Quick Assign

Quickly assign a single candidate to a PT character without opening full candidate selection. If PT/CT frequency counts do not match expected distribution, the app shows a warning modal and requires explicit confirmation.

Quick Assign

PT pattern (e.g. DOG)

CT token (e.g. 111)

Assign

Merge PT / Swap CT Tokens / Edit Cells

Advanced controls for manipulating the mapping grid.

Merge PT Cells:

- Combine two adjacent PT cells into a single cell.
- Useful when you know a plaintext pattern is a bigram/trigram.
- Also, PT groups can be specified inline using square brackets — e.g. enter [WORD] in the PT input to treat WORD as a single PT cell (maps to one CT token in single-key mode, or multiple tokens in homophone mode).
- **IMPORTANT** Merged PT cells can be splitted back when pressed *minus* button on the cell.

Swap Adjacent CT Tokens - DnD (Drag & Drop):

- Drag and drop to swap individual CT tokens.
- Only adjacent CT token swaps are allowed.
- Useful when token order appears corrupted.

Edit token text in a cell:

- Use the plus icon in a cell to open token editing and replace the current token text.

Review & Export

Review Tentative Mappings

Before exporting, review cells marked as "tentative" or unmapped:

- Verify that tentative cells align with your cryptanalytic intuition.
- Correct or lock them as needed.

Resolve Mapping Conflicts

Address any inconsistencies:

- PT characters with multiple equally-scored candidates.
- Overlapping token assignments (in multi-key mode).
- NULL cells that should be consolidated or removed.

Mark Null (Deception) Tokens

In the review phase, explicitly mark which CT tokens are **NULL** (confusing / deception):

- These will be tagged in the exported key.
- Helps document the nomenclator's structure.

Export Reconstructed Key

Click **Export Reconstructed Key** to download:

- Key solution.
The export can be saved as JSON.

Export format:

```
{
  "mode": "single",
  "key": {
    "A": "11",
    "B": "22"
  },
  "nulls": ["99"]
}
```

- mode : single or multiple
- key : PT -> CT mapping (single string in single mode, string array in multi-key mode)
- nulls (optional): deception/NULL CT tokens

Examples

Example 1: Separator mode, single-key, with deception token

This example uses your exact data:

- PT: AHAHO
- CT: 11:22:99:11:99:22:99:33:99
- Separator: :
- Keys per PT char: Single (1:1)

Expected intuition:

- 99 is deception/NULL
- Real mapping is:
 - A -> 11
 - H -> 22
 - O -> 33

Step-by-step:

1. Paste PT AHAHO and CT 11:22:99:11:99:22:99:33:99 .
2. In parsing configuration, choose Separated by character , set separator to : , and keep Single (1:1) mode.
3. Click **Run analysis** once to generate initial candidates.
4. In this dataset, o is often easy to resolve as 33 because there is one o and one 33 occurrence.
5. Open suggestions:
 - A should strongly prefer 11 or 22 .
 - H should strongly prefer 11 or 22 .
6. Try both assignments; the mapping becomes consistent when A -> 11 and H -> 22 .

PT characters: 5 CT tokens: 9

MAPPING GRID *Drag PT characters to merge , drag CT tokens to swap*

☐ Locked
 ☐ Unlocked
 ☐ Error / empty
 ☐ Highlighted

MAPPING GRID *Drag PT characters to merge , drag CT tokens to swap*

⚠ CT has more tokens than PT characters — mark extra tokens as **Null / Deception** using the panel above.

PT characters: 5 CT tokens: 9

MAPPING GRID *Drag PT characters to merge, drag CT tokens to swap*

■ Locked
 ■ Unlocked
 ■ Error / empty
 ■ Highlighted

MAPPING GRID *Drag PT characters to merge, drag CT tokens to swap*

⚠ CT has more tokens than PT characters — mark extra tokens as **Null / Deception** using the panel above.

A 11	H 22	99	A 11	99
H 22	99	0 33	99	

7. In the NULL/deception table, mark 99 as NULL; it will disappear from active mapping.
8. Review mapping table for tentative cells. In this clean case, tentative cells should usually disappear after correct deception marking.
9. Lock mappings (Lock all or Apply in suggestions), then export the reconstructed key JSON.

Typical export for this example:

```
{
  "mode": "single",
  "key": {
    "A": "11",
    "H": "22",
    "O": "33"
  },
  "nulls": ["99"]
}
```

Notes:

- If 99 is not marked as deception before final scoring, candidate quality can degrade because frequency gets polluted.
- If anything remains ambiguous, lock obvious pairs first (A->11 , H->22) and re-run analysis for the rest.

Example 2: Fixed-length with shift and NULL extraction

This example follows a practical correction workflow in fixed-length mode.

- PT: AHAHO
- CT: 1122112993399
- CT format: Fixed length
- Token length: 2
- Keys per PT char: Single (1:1)

Step-by-step:

1. Paste PT AHAHO and CT 1122112993399 .
2. Choose **Fixed length**, set token length to 2 , and click **Run analysis**.
3. In suggestions, assign A -> 11 first.
4. Move to H , then use **Shift Right** where needed to realign neighboring CT groups.
5. At O , choose 33 because you already suspect the isolated 2 is a corruption/artifact.
6. In the cell where only 2 remains for H , click the **plus** icon and edit 2 to 22 .
7. Click **Preview**. With the corrected token text, suggestions should resolve more cleanly.
8. In the NULL/deception table, mark 99 as null token.
9. Lock the mapping (Lock all, or apply and lock manually).
10. Export the reconstructed key JSON.

Expected outcome:

- A resolves to 11 .
- H resolves to 22 after shift and token correction.
- O resolves to 33 .
- 99 is exported as NULL/deception.

Example 3: Homophone mode

Use this example to observe forward lookahead and hard-locked boundaries.

- PT: ABABA
- CT: 11:22:44:11:22:55

- CT format: Separated by character
- Separator: :
- Keys per PT char: Multiple (homophones)

Step-by-step:

1. Paste PT and CT values above.
2. Choose separator mode with : and set keys per PT char to **Multiple (homophones)**.
3. Click **Run analysis**.
4. Lock B -> 22 first.
5. Then lock A -> 11 and add A -> 55 as a second homophone candidate.
6. Re-run analysis and inspect assignments near tokens not in locked sets (for example 44).
7. Notice that forward lookahead does not cross CT groups hard-locked for another PT character.
8. Resolve tentative cells manually if needed, then export.
9. Select 44 as a null.

Expected outcome:

- A can keep multiple CT tokens.
- Locked B tokens create hard boundaries that constrain lookahead.
- Tentative cells indicate places where manual review is required.

Troubleshooting

Issue: "Shift Left/Right" is greyed out.

Cause: The UI is blocking shifts into a locked neighbor.

Solution: Unlock the neighbor cell, perform the shift, then re-lock if needed.

Issue: Parsing fails or tokens are unexpectedly split.

Cause: Parse mode or separator is incorrect.

Solution:

1. Double-check the ciphertext format (fixed-width vs. delimited).
2. Re-run **Choose Parse Mode** and select the correct option.
3. If using fixed-length mode, verify token length and check status messages (`invalid` , `needsNull`).

Issue: Homophone cells are marked "tentative".

Cause: No local lookahead match or token doesn't belong to any locked candidate.

Solution: Review the cell's candidates and manually lock the correct one, or adjust neighboring locks.

Issue: Suggestions are missing (or appear unexpectedly).

Cause:

1. No lock exists yet (soft suggestions are intentionally disabled until at least one lock is set).
2. The PT character is in an ambiguous locked segment (NULL/deception or tentative cells between lock anchors).

Solution:

1. Set at least one reliable lock first.
2. Resolve NULL/deception and tentative cells in the surrounding interval, or some other PT has wrong cipher marked.

Issue: Shift is disabled even though I need to rebalance groups.

Cause: Shift can be blocked by a locked immediate neighbor or by edge/cascade constraints.

Solution:

1. Unlock neighboring locked mappings temporarily.
2. Retry shift from a cell that has more than one token.
3. If near first/last cell, rebalance from inner cells first, then lock again.

Issue: Export contains unexpected NULL tokens.

Cause: Tokens were not explicitly mapped or were marked as deception.

Solution: Before export, review the **Mark Null (Deception)** list and confirm which tokens should be tagged as NULL.

Issue not known

- Return to a previous stable state, or restart from the beginning and avoid repeating the step that introduced the issue.

Appendix: Glossary

Term	Definition
CT Token(Ciphertext Token)	A unit of encrypted text (character in fixed-length, or delimiter-separated part in separator mode).
PT (Plaintext)	The original message before encryption.
NULL	A deception token with no corresponding plaintext character.
Homophone	Multiple ciphertext tokens that represent the same plaintext character.
Local Lookahead	In homophone mode, a forward scan to find candidate tokens matching locked mappings; bounded by excess CT groups and stopped by hard-locked boundaries.
Suggestion (Soft)	A visual hint for likely candidates. Suggestions require at least one lock and are suppressed in ambiguous locked segments.
Tentative	A cell or mapping that is not yet confirmed or locked.
Hard-Locked Boundary	A confirmed (locked) token that halts the local lookahead scan.
maxLen	Maximum number of CT tokens per PT cell in fixed-length mode.
Group Size	In fixed-length mode, the number of characters per token. In separator mode, not applicable.

Last Updated: March 30 2026

Version: 1.4